

Tr	Part Name	 Subsystem	 Procurement	Official Description	Tr	Link	 Vendor	Part Number	Qty	Length	Unit Cost	System Cost	Order Cost	Notes
	Angle Bracket Assembly	Roller	Purchased	T-Slotted Framing Gusset Bracket for 20mm High Rail, 1-5/8" Long, Silver		https://www.mcmaster.com/5537T658	McMaster Carr	5537T658	2		\$9.30	\$18.60	\$18.60	
	Antislip Leveling Mount for 20 mm High Single Rail	Feed	Purchased	Antislip Leveling Mount for 20 mm High Single Rail		https://www.mcmaster.com/5537T982	McMaster Carr	5537T982	8		\$6.74	\$53.92	\$53.92	
	Base Plate	Roller	Fabricated	Multipurpose 6061 Aluminum Bar		https://www.mcmaster.com/8975K437	McMaster Carr	8975K437	1		\$23.17	\$23.17	\$23.17	alternative (41.69 mic 6 tooling plate https://www.mcmaster.com/86825K722/)
	Bearing (Top Rollar)	Feed	Purchased	608 ZZ Ball Bearings(10PCS), 608ZZ Metal Double Shielded Miniature Deep Groove Skateboard Ball Bearings (8mm x 22mm x 7mm)		https://www.amazon.com/dp/B078888888	Amazon		6		\$0.59	\$3.53	\$5.88	Sold As Pack Of 10
	Bearing Block	Roller	Fabricated	Bar Made From Teflon® PTFE 1/2" Thick, 1" Wide		https://www.mcmaster.com/8735K37	McMaster Carr	8735K37	1	3	\$38.49	\$6.42	\$38.49	Buy shortest (i.e.1 ft) only need 3 inches
	Corner Bracket Assembly	Feed, Roller	Purchased	T-Slotted Framing Silver 90 Degree Surface Bracket for 20 mm High Rail		https://www.mcmaster.com/5537T962	McMaster Carr	5537T962	10		\$11.08	\$110.80	\$110.80	
	DC Supply	Roller	Purchased	AC/DC CONVERTER 12V 36W		https://www.digikey.com/product-detail/en/1866-3338-ND	DigiKey	1866-3338-ND	1		\$10.80	\$10.80	\$10.80	
	Electrical Housing	Roller	Purchased	Zulkit Junction Box Blue Metal Project Box DIY Electric Enclosure Case Preventive Case Electrical Box 7.9 x 6.5 x 3.5 inch		https://www.amazon.com/dp/B078888888	Amazon		1		\$19.99	\$19.99	\$19.99	Assumed speed controller on top of DC converter
	End Cap for 20 mm High Single Rail	Feed	Purchased	End Cap for 20 mm High Single Rail, Black		https://www.mcmaster.com/5537T14	McMaster Carr	5537T14	16		\$1.06	\$16.96	\$16.96	
	End Cap for 40 mm High Double Rail	Feed	Purchased	End Cap for 40 mm High Double Rail, Black		https://www.mcmaster.com/5537T19	McMaster Carr	5537T19	4		\$1.56	\$6.24	\$6.24	
	Feed End Bar	Feed	Purchased	Single Four Slot Rail, Silver, 20 mm Square, Solid		https://www.mcmaster.com/5537T101	McMaster Carr	5537T101	2	1ft	\$6.52	\$13.04	\$13.04	
	Feed Length Bar	Feed	Purchased	Single Four Slot Rail, Silver, 20 mm Square, Solid		https://www.mcmaster.com/5537T101	McMaster Carr	5537T101	4	1ft	\$6.52	\$26.08	\$26.08	
	Feed Plate	Feed	Purchased	Multipurpose 6061 Aluminum Bar 1/8" Thick x 1-1/2" Wide		https://www.mcmaster.com/8975K581	McMaster Carr	8975K581	2	3ft	\$8.64	\$17.28	\$17.28	
	Feed Support Bar	Feed	Purchased	T-Slotted Framing Single Four Slot Rail, Silver, 20 mm Square, Solid		https://www.mcmaster.com/5537T911	McMaster Carr	5537T911	8	3.5in	\$2.24	\$17.92	\$17.92	
	Flexible Coupling	Roller	Purchased	Upgrade D20*L25 Flexible Shaft Coupling 8mm to 8mm Aluminium Plum Connector Flexible Coupler for 3D Printer CNC Machine and Servo Stepped Motor		https://www.amazon.com/dp/B078888888	Amazon		1		\$11.04	\$11.04	\$11.04	For Our 8mm Axel
	Hex socket countersunk head screw M5x0.8 x 35	Roller	Purchased	Zinc-Plated Alloy Steel Hex Drive Flat Head Screw M5 x 0.8 mm Thread Size, 35 mm Long		https://www.mcmaster.com/91263A883	McMaster Carr	91263A883	4		\$0.32	\$1.29	\$16.09	
	Hex socket countersunk head screw M5x0.8 x 8	Feed	Purchased	Zinc-Plated Alloy Steel Hex Drive Flat Head Screw M5 x 0.8 mm Thread Size, 8 mm Long		https://www.mcmaster.com/91263A831	McMaster Carr	91263A831	16		\$0.09	\$1.51	\$9.43	
	Hex socket countersunk head screw M5x0.8 x 8	Feed	Purchased	Zinc-Plated Alloy Steel Hex Drive Flat Head Screw M5 x 0.8 mm Thread Size, 8 mm Long		https://www.mcmaster.com/91263A831	McMaster Carr	91263A831	16		\$0.09	\$1.51	\$9.43	
	Hex socket head cap screw M4x0.70 x 8	Jig	Purchased	Super-Corrosion-Resistant 316 Stainless Steel Socket Head Screw M4 x 0.7 mm Thread, 8 mm Long		https://www.mcmaster.com/92290A140	McMaster Carr	92290A140	4		\$0.40	\$1.60	\$9.98	pack of 25
	Hex socket head cap screw M5x0.80 x 30 x 22	Roller	Purchased	Zinc-Plated Alloy Steel Socket Head Screw M5 x 0.8 mm Thread, 30 mm Long		https://www.mcmaster.com/90128A256	McMaster Carr	90128A256	4		\$0.15	\$0.61	\$15.13	
	Hex socket head cap screw M5x0.80 x 8	Roller	Purchased	Zinc-Plated Alloy Steel Socket Head Screw M5 x 0.8 mm Thread, 8 mm Long		https://www.mcmaster.com/90128A232	McMaster Carr	90128A232	4		\$0.11	\$0.42	\$10.55	
	Housing (Top Roller)	Feed	Fabricated	PETG HF		https://us.storify.com/Other	Other		2		\$22.99	\$1.84	\$22.99	40g
	IEC Cord	Roller	Purchased	ZYAMY 4PCS IEC320 Installation Inlet Module Plug C14 Power Connector Socket AC 250V 10A 3-pin Panel Mount Adapter for Computer Host Monitor Socket Fuse Holder		https://www.amazon.com/dp/B078888888	Amazon		1		\$1.97	\$1.97	\$7.89	Need to determine type of fuse
	Jig Center Overhang Shim	Jig	Fabricated	18-8 Stainless Steel Shim Stock		https://www.mcmaster.com/9784K616	McMaster Carr	9784K616	1	1in	\$1.98	\$1.98	\$0.00	Uses the same stock as Jif Side Bracing
	Jig Side Bracing	Jig	Fabricated	18-8 Stainless Steel Shim Stock		https://www.mcmaster.com/9784K616	McMaster Carr	9784K616	4	0.25in	\$0.50	\$2.00	\$11.88	
	Jig Side Overhang Shim	Jig	Fabricated	18-8 Stainless Steel Shim Stock		https://www.mcmaster.com/9784K616	McMaster Carr	9784K616	2	0.5in	\$0.99	\$1.98	\$0.00	Uses the same stock as Jif Side Bracing
	Lamina Jig Half 1	Jig	Fabricated	PLA Basic		https://us.storify.com/Other	Other		1		\$0.20	\$0.20	\$0.00	
	Lamina Jig Half 2	Jig	Fabricated	PLA Basic		https://us.storify.com/Other	Other		1		\$0.20	\$0.20	\$0.00	
	Long Zinc-Plated Alloy Steel Hex Drive Flat Head Screw	Feed	Purchased	Zinc-Plated Alloy Steel Hex Drive Flat Head Screw		https://www.mcmaster.com/91263A883	McMaster Carr	91263A883	1		\$0.32	\$0.32	\$16.09	
	M8 Medium-Strength Steel Serrated Flange Nut	Feed	Purchased	Class 8, Zinc-Plated, M8 x 1.25 mm Thread		https://www.mcmaster.com/96194A102	McMaster Carr	96194A102	24		\$0.15	\$3.55	\$14.81	

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	Medium-Strength Steel Serrated Flange Nut	Feed	Purchased	Medium-Strength Steel Serrated Flange Nut		https://www.mcmaster.com	McMaster Carr	96194A107	1		\$0.13	\$0.13	\$12.69	
	Metal weight	Jig	Purchased	Tight-Tolerance Oil-Hardening O1 Tool Steel Bar		https://www.mcmaster.com	McMaster Carr	9516K542	1	1.5 ft	\$148.22	\$148.22	\$148.22	make sure its the 1-1/2 ft bar 1/4 in. thick this is to weight down our jig for stacking
	Motor	Roller	Purchased	BRINGSMART 12V 18rpm DC Worm Gear Motor 12kg.cm Low Speed Mini Turbine Worm 12V Reducer Electric Motor Self-Locking for Toys (JGY-370 12V 18rpm)		https://www.amazon.com	Amazon		1		\$15.87	\$15.87	\$15.87	18 RPM
	Motor Bracket	Roller	Fabricated	Bringsmart 370 Mounting Bracket with Screw Worm Gear Motor Fixed Fastener DIY Parts DC Motor Holder (Single-Side Bracket)		https://www.amazon.com	Amazon		1		\$8.99	\$8.99	\$8.99	
	Motor Mount Spacer	Feed	Fabricated	PETG HF		https://us.storify.com	Other		1		\$20.00	\$0.16	\$0.00	
	MPHX Jig Half 1	Jig	Fabricated	PLA Basic		https://us.storify.com	Other		1		\$0.20	\$0.20	\$0.00	
	MPHX Jig Half 2	Jig	Fabricated	PLA Basic		https://us.storify.com	Other		1		\$0.20	\$0.20	\$0.00	
	Pan	Roller	Purchased	Custom Machined Vat		https://www.printables.com	Other		1		\$300.00	\$300.00	\$300.00	Custom
	Pan Cover	Roller	Purchased	Goodma 2Pcs Rectangle Metal Tins Containers with Window Lids Mini Storage Canister Jars Set for Home Kitchen, Office and School Supplies, 8.8" x 4.5" x 3.5" (Black)		https://www.amazon.com	Amazon		1		\$9.00	\$9.00	\$17.99	
	Pivot Assembly	Roller	Purchased	T-Slotted Framing Inline/Perpendicular Pivot for 20 mm Single Rail		https://www.mcmaster.com	McMaster Carr	5537T219	2		\$21.77	\$43.54	\$43.54	
	Plain washer large grade A M5	Roller	Purchased	Zinc-Plated Steel Oversized Washer for M5 Screw, 5.3 mm ID, 15 mm OD, 1 mm to 1.4 mm Thick		https://www.mcmaster.com	McMaster Carr	91100A140	2		\$0.05	\$0.10	\$5.12	
	Plain washer normal grade A M8	Feed	Purchased	Zinc-Plated Steel General Purpose Washer for M8 Screw Size, 8.4 mm ID, 16 mm OD		https://www.mcmaster.com	McMaster Carr	91166A270	24		\$0.03	\$0.80	\$3.32	
	Roller	Roller	Fabricated	Multipurpose 6061 Aluminum Rod		https://www.mcmaster.com	McMaster Carr	8974K18	1		\$42.23	\$9.15	\$42.23	For a unit we need 5.2 inch but we want to buy 2 ft to be safe and test kurling
	Roller Axle	Roller	Fabricated	Tight-Tolerance Low-Carbon Steel Rod		https://www.mcmaster.com	McMaster Carr	6103N46	1	12.038	\$21.56	\$7.19	\$21.56	Buy 3ft we only need a 12.038 inches for the unit cost but we need to buy 3 ft
	Roller Back Vertical Bar	Feed	Purchased	T-Slotted Framing Single Four Slot Rail, Silver, 20 mm Square, Solid		https://www.mcmaster.com	McMaster Carr	5537T911	2	4.425in	\$2.83	\$5.66	\$5.66	
	Roller Cross Bar	Feed	Purchased	T-Slotted Framing Single Four Slot Rail, Silver, 20 mm Square, Solid		https://www.mcmaster.com	McMaster Carr	5537T911	1	4.425in	\$2.83	\$2.83	\$2.83	
	Roller End Double Bar	Feed	Purchased	T-Slotted Framing Double Six Slot Rail, Silver, 40 mm High x 20 mm Wide, Solid		https://www.mcmaster.com	McMaster Carr	5537T111	2	1ft	\$8.77	\$17.54	\$17.54	
	Self-Aligning T-Slotted Framing Nut	Feed, Roller	Purchased	Self-Aligning T-Slotted Framing Nuts with Spring-Loaded Ball, Drop-in, for 20mm Rail, M5 Thread		https://www.mcmaster.com	McMaster Carr	5278N44	60		\$1.27	\$76.35	\$76.35	
	Set Screw	Roller	Purchased	18-8 Stainless Steel Cup-Point Set Screw		https://www.mcmaster.com	McMaster Carr	92311A190	1		\$0.07	\$0.07	\$6.77	Sold As Pack of 100
	Shaft (Top Roller)	Feed	Fabricated	Tight-Tolerance Low-Carbon Steel Rod		https://www.mcmaster.com	McMaster Carr	6103N46	1		\$21.56	\$1.80	\$0.00	Same Stock As Axel For Adhessive Roller (can be cut from same piece)
	Shaft Collar	Roller	Purchased	303 Stainless Steel Set Screw Shaft Collar		https://www.mcmaster.com	McMaster Carr	6058N16	2		\$5.68	\$11.36	\$11.36	Sold Separly
	Socket button head screw M3x0.5 x 8	Jig	Purchased	Button Head Hex Drive Screw Passivated 18-8 Stainless Steel, M3 x 0.50 mm Thread, 8mm Long		https://www.mcmaster.com	McMaster Carr	92095A181	16		\$0.09	\$1.41	\$8.80	
	Socket button head screw M5x0.8 x 12	Roller	Purchased	Button Head Hex Drive Screws, Zinc-Plated Alloy Steel, M5 x 0.8 mm Thread, 12 mm Long		https://www.mcmaster.com	McMaster Carr	91306A666	8		\$0.13	\$1.06	\$13.20	
	Socket button head screw M5x0.8 x 16	Roller	Purchased	Button Head Hex Drive Screws Zinc-Plated Alloy Steel, M5 x 0.8 mm Thread, 16 mm Long		https://www.mcmaster.com	McMaster Carr	91306A667	6		\$0.14	\$0.84	\$14.03	
	Socket button head screw M8x1.25 x 22	Feed	Purchased	Button Head Hex Drive Screws Zinc-Plated Alloy Steel, M8 x 1.25 mm Thread, 20 mm Long		https://www.mcmaster.com	McMaster Carr	91306A678	24		\$0.37	\$8.84	\$9.21	
	Speed Contoller	Roller	Purchased	Bringsmart CCM6N DC PWM Motor Speed Controller 6V-30V Variable Speed Drive 6A Positive Inversion Mini Switch Controller (Speed Controller)		https://www.amazon.com	Amazon		1		\$14.27	\$14.27	\$14.27	Product Specs
	Spring (Top Roller)	Feed	Purchased	W-100CS - 1.19 Inch 302/304 Stainless Steel Compression Springs		https://www.cscspring.com	Other		4		\$7.34	\$29.36	\$29.36	
	Sqeegee	Roller	Fabricated	Polished Multipurpose 304 Stainless Steel Strip		https://www.mcmaster.com	McMaster Carr	9085K11	1	7	\$19.49	\$11.37	\$19.49	order 1ft only need 7 inch

Tt	Part Name	🔗 Subsystem	🔗 Procurement	Offical Description	Tt Link	🔗 Vendor	Part Number	Qty	Length	Unit Cost	System Cost	Order Cost	Notes
	Super Glue for Securing Overhang Plates	Jig	Purchased	3 Gram (Pack of 24) Single Use Super Glue Liquid, Super Fast, Thin & Strong Adhesive Superglue, Cyanoacrylate Glue for Hard Plastics, DIY Craft, Ceramics, Leather, Metal Etc	https://www.amazon.com/gp/product/B00006BI66	Amazon		1		\$0.34	\$0.34	\$8.09	
	T-Bracket Assembly	Feed, Roller	Purchased	T-Slotted Framing Silver Tee Surface Bracket for 20mm High Single Rail	https://www.mcmaster.com/91746A648	McMaster Carr	5537T971	9		\$13.31	\$119.79	\$119.79	
	Thumb Screw	Roller	Purchased	Knurled-Head Thumb Screw	https://www.mcmaster.com/91746A648	McMaster Carr	91746A648	3		\$3.96	\$11.88	\$11.88	Uesd For Squeegee (only need 3) (sold as pack fo 10)
	Top Plate	Feed	Fabricated		https://us.storify.com/other	Other		2		\$0.22	\$0.43	\$0.00	
	Top Roller Housing	Feed	Fabricated	PETG HF	https://us.storify.com/other	Other		2		\$20.00	\$8.00	\$19.99	
	Top Roller Latch Bar	Roller	Purchased	Single Four Slot Rail, Silver, 20 mm Square, Solid	https://www.mcmaster.com/5537T101	McMaster Carr	5537T101	1	1/2ft	\$9.47	\$9.47	\$9.47	
	Top Roller Mounting Bar	Feed	Purchased	T-Slotted Framing Single Four Slot Rail, Silver, 20 mm Square, Solid	https://www.mcmaster.com/5537T911	McMaster Carr	5537T911	2	10.425in	\$6.67	\$13.34	\$13.34	
	Washer	Roller	Purchased	18-8 Stainless Steel Washer for Number 10 Screw, General Purpose, 0.203" ID, 0.438" OD	https://www.mcmaster.com/92141A011	McMaster Carr	92141A011	2		\$0.02	\$0.05	\$2.40	Motor Mount Washer (need 2) (pack of 100)
	Wire	Roller	Purchased	18 AWG Silicone Wire Kit 50FT, 18 Gauge Stranded Tinned Copper Wire Spool, 5 Colors 10FT Each, Flexible Electrical Wire for DIY, Automotive, LED, Arduino, Battery Projects	https://www.amazon.com/gp/product/B00006BI66	Amazon		1	24	\$0.90	\$0.90	\$8.99	10 ft assorted color wire